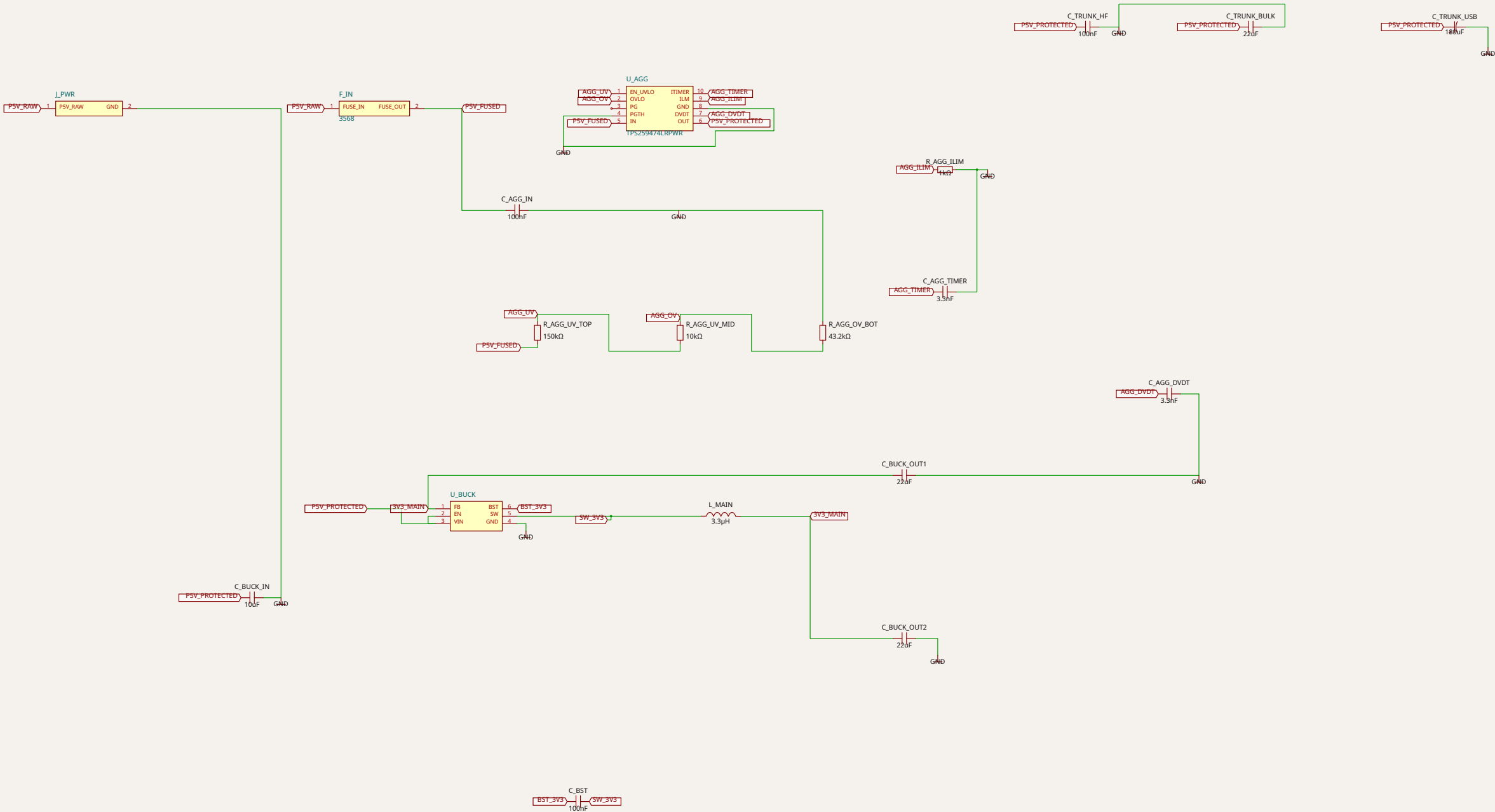


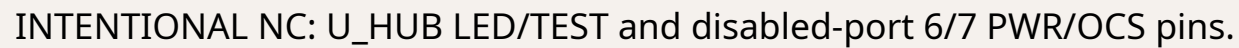
USB_CONTROLLED_DEBUG_HUB

POWER — regulated 5 V input, fuse, aggregate eFuse and 3.3 V buck

Page 1 of 9 • exact circuit.json SHA-256 ff20fe16fd350379... • 19 components • landscape page fit

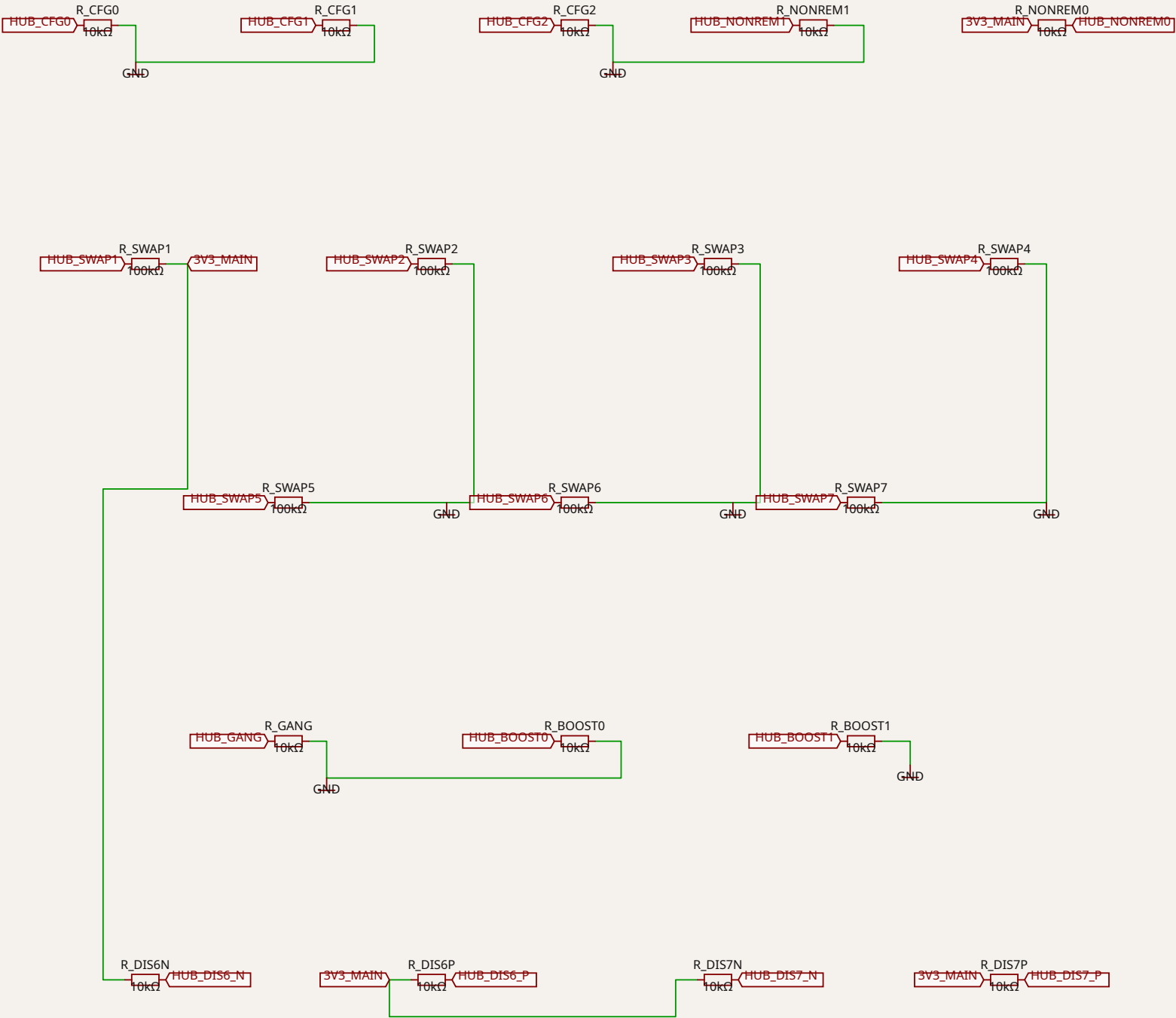


USB HUB — upstream Type-B, ESD, USB2517I, straps, clock and bypass



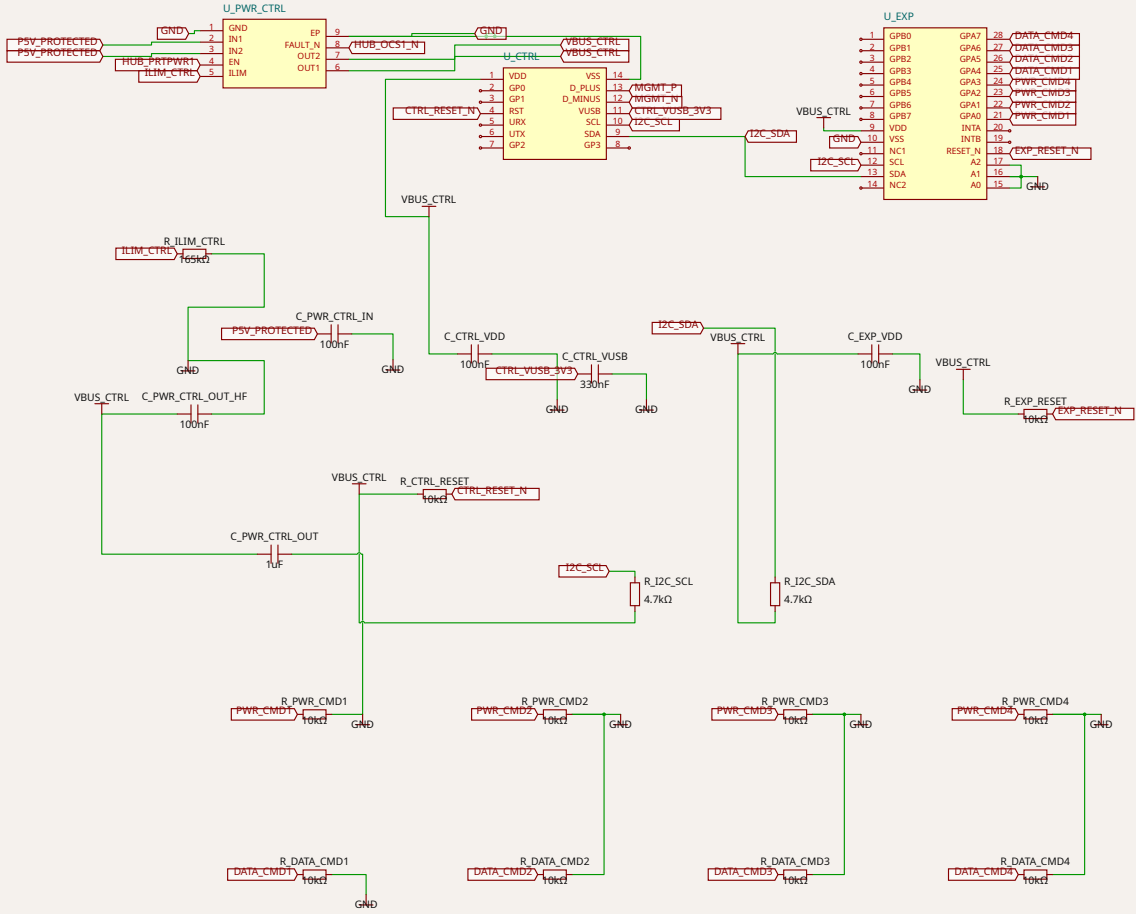
USB_CONTROLLED_DEBUG_HUB

USB HUB CONFIGURATION — P1 swapped, P2-5 normal, P6-7 disabled



MANAGEMENT — factory MCP2221A HID/I2C and MCP23017 command bank

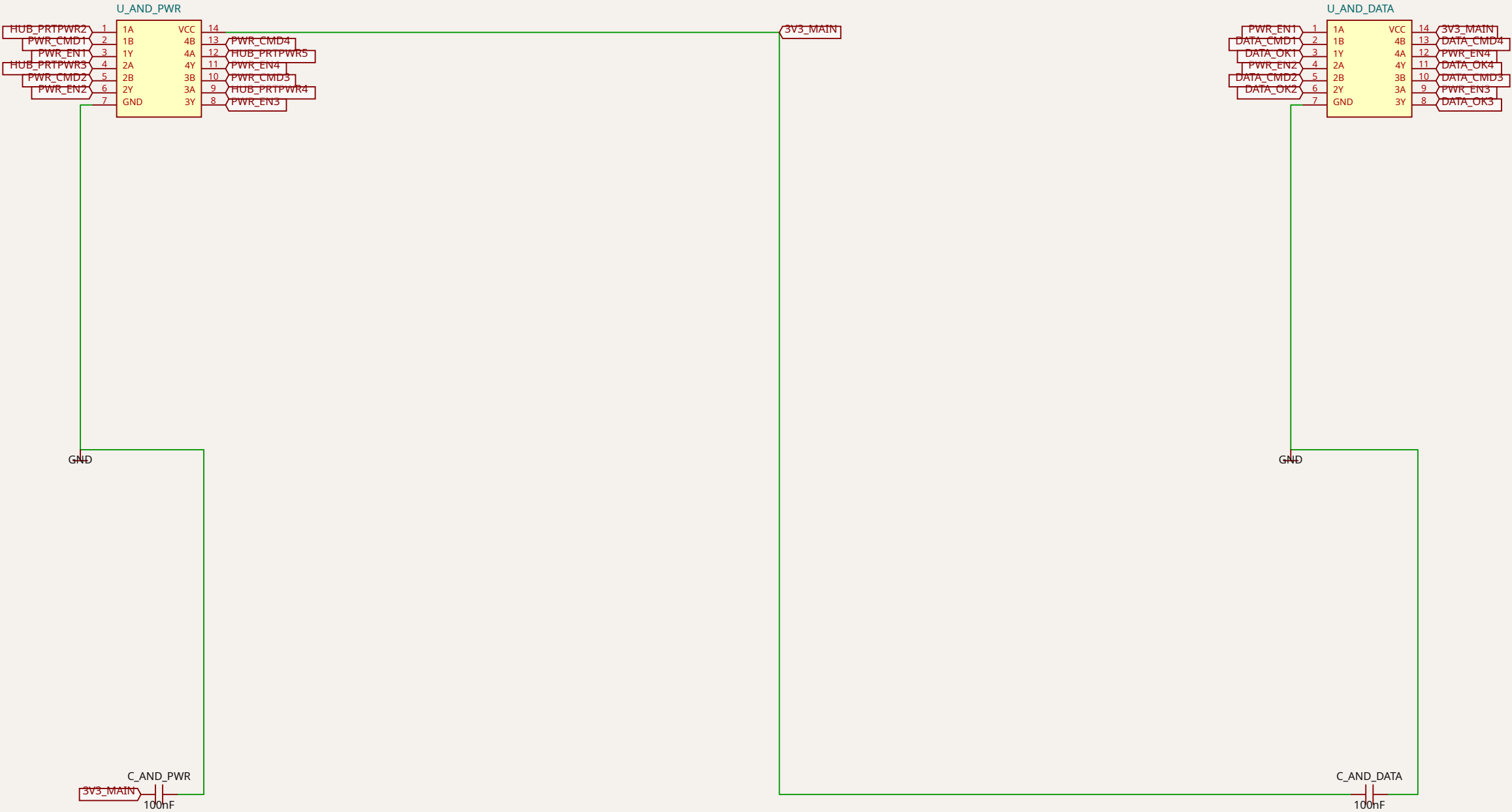
Page 4 of 9 • exact circuit.json SHA-256 ff20fe16fd350379... • 22 components • landscape page fit



INTENTIONAL NC: unused U_CTRL GPIO/UART; U_EXP GPB/INT/NC pins.

USB_CONTROLLED_DEBUG_HUB

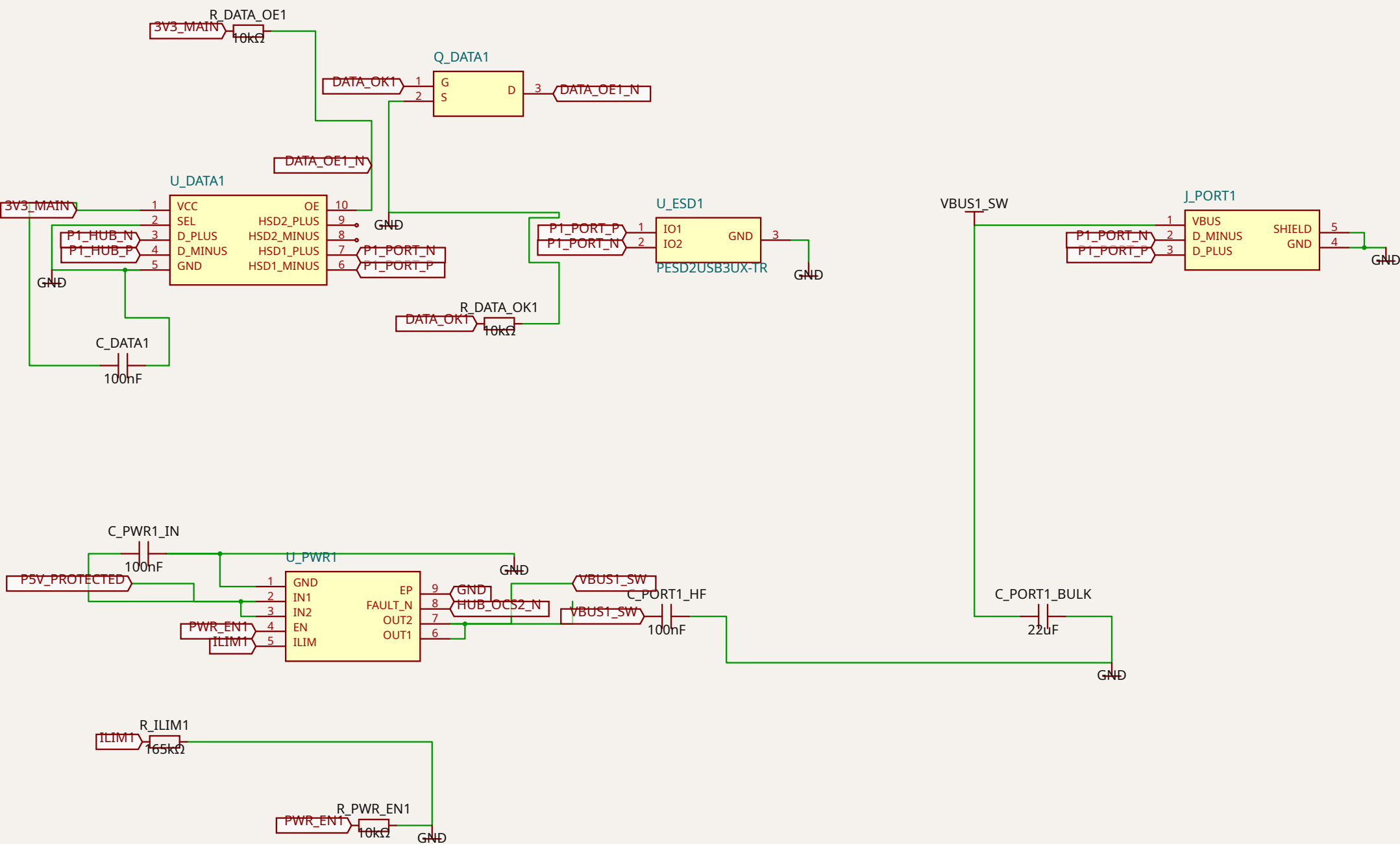
INTERLOCKS — hub policy AND host command; data follows commanded power enable



USB_CONTROLLED_DEBUG_HUB

EXTERNAL PORT 1 — independent power/data disconnect

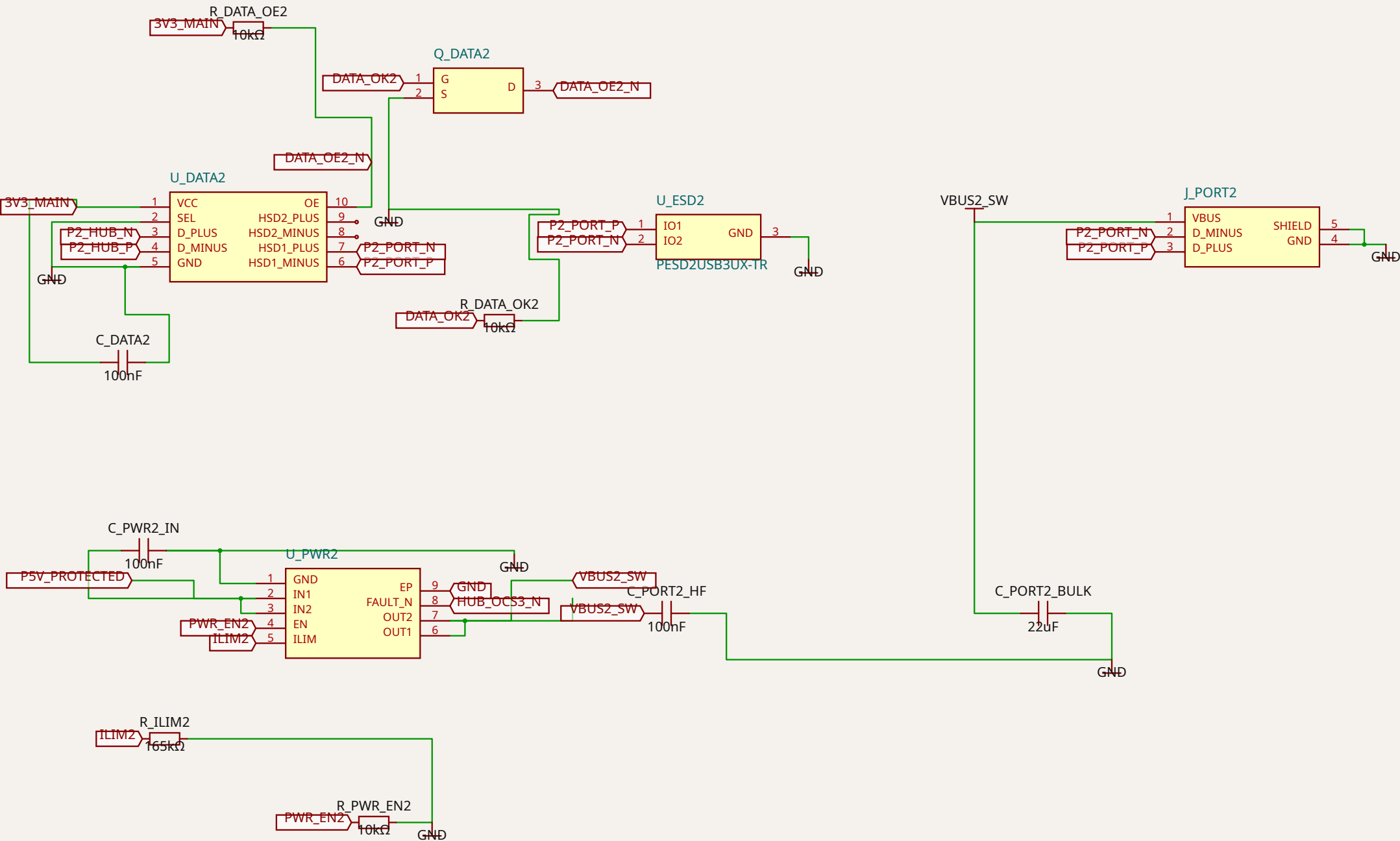
Page 6 of 9 • exact circuit.json SHA-256 ff20fe16fd350379... • 13 components • landscape page fit



USB_CONTROLLED_DEBUG_HUB

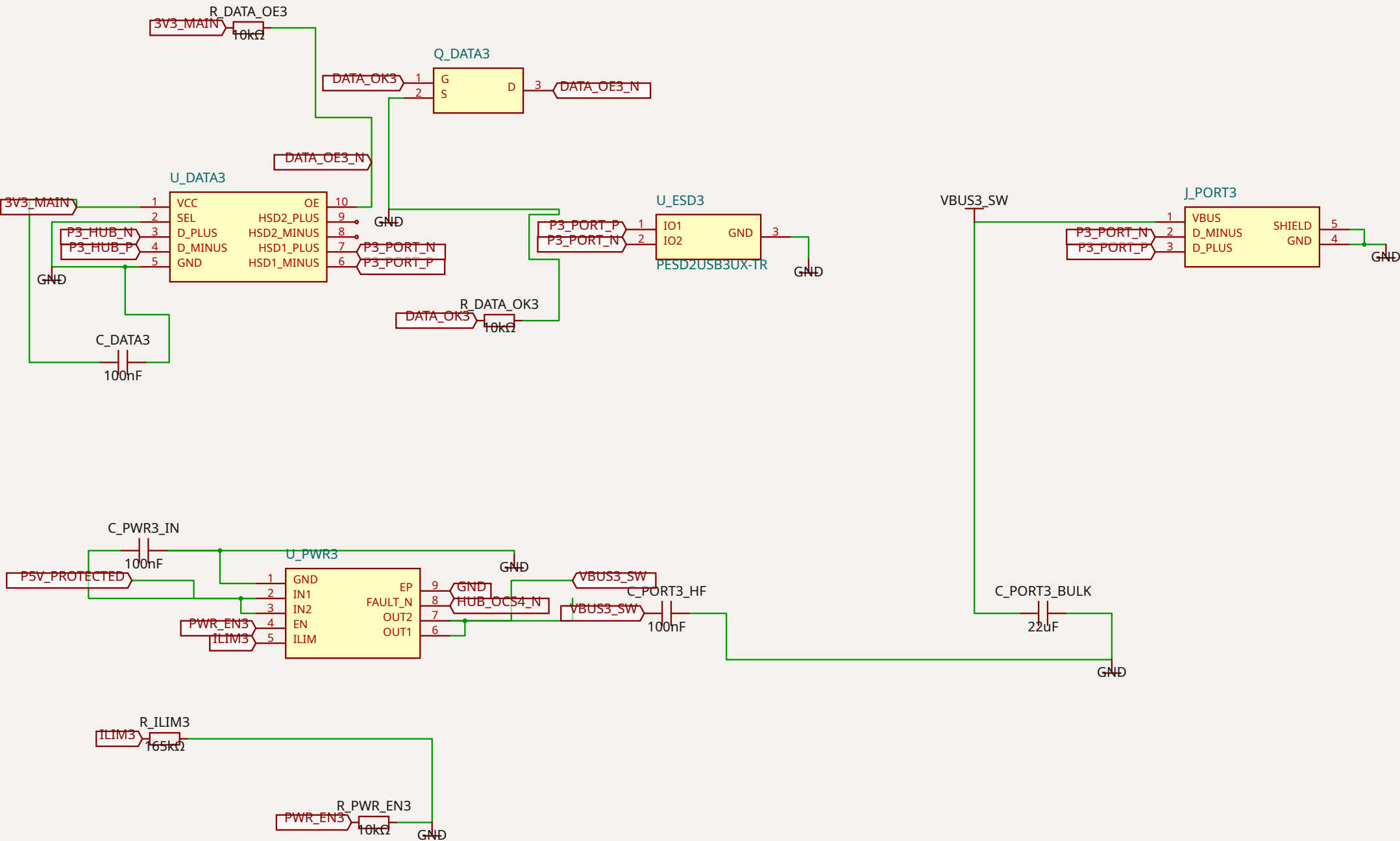
EXTERNAL PORT 2 — independent power/data disconnect

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USB_CONTROLLED_DEBUG_HUB

EXTERNAL PORT 3 — independent power/data disconnect



USB_CONTROLLED_DEBUG_HUB

EXTERNAL PORT 4 — independent power/data disconnect

